Afrouz Delshad

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RESEARCH INTERESTS

Machine learning for time-series forecasting (Recurrent Neural Networks, Transformers), LLM evaluation, LLM-as-a-judge.

EDUCATION

Georgia Institute of Technology, PhD, Computational Science and Engineering — Expected May 2029
Advisor: Prof. Elizabeth M. Cherry

Georgia Institute of Technology, MS, Computational Science and Engineering — May 2026

University of Kashan, BS, Applied Mathematics — Feb 2021

EXPERIENCE

Graduate Research Assistant, Georgia Institute of Technology — May 2024 – Present
LLM judges for scientific time series

- Designed a controlled benchmark in which complexity and chaos of time series vary independently with known ground truth, to test whether LLM judges measure the property they are asked to evaluate.
- Engineered task prompts and evaluated Claude, GPT and Gemini on four tasks: ranking, outlier detection, clustering, and pairwise evaluation.
- Analyzed model reasoning and realized that failures usually came from plausible but incorrect proxy features.

Deep and hybrid echo state networks for forecasting

- Studied echo state networks (ESNs), recurrent neural networks with a fixed random reservoir in which only the output layer is trained, for forecasting chaotic and real-world time series.
- Developed six deep ESN architectures and hybrid versions that integrate physics-based model data into the ML model.
- Improved accuracy by up to 65% on chaotic Mackey–Glass data and 14% on zebrafish cardiac recordings over baseline ESNs with the same total number of neurons, considering the effect of depth rather than network size.
- Tuned hyperparameters with Bayesian optimization and showed that deep architectures make forecasts more robust.

Transformers for complex time-series prediction

- Applied Transformers with different attention mechanisms (self-attention, easy attention) to forecast complex time series.

Graduate Teaching Assistant, Georgia Institute of Technology — Aug 2026 – Present
Biomedical Modeling and Simulation (CSE 8803)

- Helped design course assignments and delivered lectures on selected topics.

PUBLICATIONS AND PRESENTATIONS

1. **Delshad A**, An X, Fenton FH, Cherry EM. Do LLM judges evaluate what we ask them to measure? A controlled benchmark for complexity and chaos in scientific time series. Manuscript, 2026. PDF
2. **Delshad A**, Cherry EM. Predicting complex time series with deep echo state networks. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 35(9), 2025. https://doi.org/10.1063/5.0283425
3. **Delshad A**, Cherry EM. Deep echo state networks improve cardiac voltage time series prediction. Talk, *SIAM Conference on the Life Sciences*, July 2026.
4. **Delshad A**, Cairns D, Cherry EM. Efficacy of hybrid echo state networks in forecasting cardiac electrophysiological signals. Poster, *SIAM Conference on Mathematics of Data Science*, Oct 2024.

AWARDS

Student Travel Award, SIAM Conference on the Life Sciences — 2026

SKILLS

Programming: Python (NumPy, SciPy, pandas, scikit-learn, PyTorch, Matplotlib), MATLAB, Git, LaTeX

Machine Learning: Transformers, Recurrent Neural Networks (RNNs), Reservoir Computing, Physics-informed Neural Networks (PINNs), Bayesian Optimization

LLMs & Evaluation: LLM-as-a-judge, benchmark design, prompt engineering, reasoning analysis